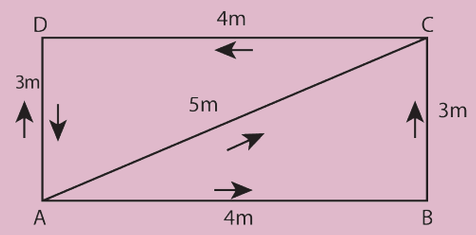
Activity 3

Observe the motion of a car as shown in the figure and answer the following questions:



Compare the distance covered by the car through the path ABC and AC. What do you observe? Which path gives the shortest distance to reach D from A? Is it the path ABCD or the path ACD or the path AD?

2.4 Speed, Velocity and Acceleration

Speed is the quantity which shows how fast the body is moving but velocity is the quantity which shows the speed as well as the direction of the moving body.

2.4.1 Speed

Speed is the rate of change of distance or the distance travelled in unit time. It is a scalar quantity. The SI unit of speed is ms^{-1}.
Speed = Distance travelled / Time taken

Problem 1

An object travels 16 m in 4 s and then another 16 m in 2 s. What is the average speed of the object?

Solution:

Total distance travelled by the object = 16 m + 16 m = 32 m

Total time taken = 4s + 2s = 6s

$$\text{Average speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}} = \frac{32m}{6s} = 5.33\ ms^{-1}$$

Therefore, the average speed of the object is 5.33 ms^{-1}

Problem 2

A sound is heard 5 s later than the lightning is seen in the sky on a rainy day. Find the distance of location of lightning. Given the speed of sound = 346 ms^{-1}

Solution:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Distance = Speed × Time = 346 × 5 = 1730 m

Thus, the distance of location of lightning is 1730 m.

2.4.2 Velocity

Velocity is the rate of change of displacement. It is the displacement in unit time. It is a vector quantity. The SI unit of velocity is ms^{-1}.



Velocity = Displacement / Time taken

2.4.3 Acceleration

Acceleration is the rate of change of velocity or it is the change of velocity in unit time. It is a vector quantity. The SI unit of acceleration is ms^{-2}.

Acceleration
= Change in velocity/Time
= (Final velocity – Initial velocity)/Time
$a = (v-u)/t$

Consider a situation in which a body moves in a straight line without reversing its direction. From the above equation if $v > u$, i.e. if final velocity is greater than initial velocity, the velocity increases with time and the value of acceleration is positive.

If $v < u$, i.e. if final velocity is less than initial velocity, the velocity decreases with time and the value of acceleration is negative. It is called negative acceleration. Negative acceleration is called retardation or deceleration. If the acceleration has a value of $-2\ ms^{-2}$, we say that deceleration is $2\ ms^{-2}$.

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